

Patricio Antonio García Pérez Vela

Software Engineer | Full-Stack, AI & Data Systems

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Profile

Recent Computer Systems Engineering graduate focused on full-stack products, automation, applied AI, and data systems. At Mazda, I built an IT document-management system that replaces Excel/email-based controls with role-based workflows, validations, traceability, and SQL Server revision history. Main stack: Python/FastAPI, TypeScript/Next.js, C#/.NET, PostgreSQL/pgvector, Docker, and NLP/voice models.

Technical Skills

Full-stack: TypeScript, JavaScript, React, Next.js, Node.js, Express, FastAPI, REST APIs, HTML/CSS.

AI, data & ML: Python, Pandas, scikit-learn, spaCy, TensorFlow, Whisper/faster-whisper, Gemini/OpenAI APIs, embeddings, pgvector, NLP, computer vision.

Databases: PostgreSQL, SQL Server, MongoDB, Supabase, Entity Framework, Alembic.

Cloud, DevOps & tools: Docker, GitHub Actions, Vercel, Render, Fly.io, Git, Linux, WSL2, Scrum/SDLC.

Enterprise: C#/.NET, Windows Forms, Active Directory, SharePoint. **Languages:** Spanish native, English B2.

Professional Experience

Mazda Motor Manufacturing de México

Salamanca, Guanajuato, Mexico

Software Engineer Intern – IT Document Management Digitalization FY25

Aug 2025 – Feb 2026

- Transformed an Excel, SharePoint, and email-based document process into a C#/.NET desktop application with SQL Server for creation, updates, obsolescence, search, and traceability.
- Modeled the relational database and Entity Framework data access for review requests, document states, traceability numbers, and auditable revision history.
- Implemented Active Directory authentication, roles, and permissions to restrict critical operations and reduce the risk of unauthorized edits or deletion.
- Designed forms, validations, and approval workflows with projected improvements: metadata updates from 20 to 3 minutes, automatic traceability generation after review, and prevention of duplicate/corrupted folios.

Stack: C#/.NET, Windows Forms, SQL Server, Entity Framework, Active Directory, SharePoint, Krypton Toolkit, SDLC

Selected Projects

DEMOX – Political Intelligence Platform

2026

Private repository | Full-stack + AI

- Built a political media intelligence platform using Next.js and FastAPI. Implemented interactive party trend dashboards, data freshness alerts, evidence-backed reports, and background processing with Celery/Redis.

Stack: Python, FastAPI, Next.js, PostgreSQL/Supabase, pgvector, Celery, Redis, spaCy, Gemini, Docker

WhisperKey – 100% Local Voice Dictation

2026

github.com/p5Patricio/WhisperKey

- Developed a bilingual offline voice dictation application transcribing in real-time on GPU using Whisper. Features automatic hardware detection, interactive setup wizard, and a CustomTkinter graphical installer.

Stack: Python, Whisper, faster-whisper, CustomTkinter, CUDA, PyInstaller, hotkeys, cross-platform

NBA Player Profiling – Sports Machine Learning

2025

GitHub

- Restructured into a monorepo with a FastAPI backend and a Next.js frontend (Tailwind v4 & Chart.js). Clusters players via K-Means using official NBA API statistics and generates interactive dashboards and PDF reports.

Stack: Python, FastAPI, Next.js (TypeScript), scikit-learn, Pandas, NBA API, Chart.js, Tailwind v4

Fitodex – Agrochemical Control for DICIVA

2024 – 2025

Universidad de Guanajuato project | Private repository

- Developed a REST API to digitize agrochemical catalog and control workflows: insecticides, crops, pests, dosage, technical sheets, safety documents, users, roles, and password recovery.

Stack: Node.js, Express, MongoDB Atlas, Mongoose, JWT, bcrypt, Nodemailer, Docker, Fly.io

Education

Universidad de Guanajuato (DICIS)

Salamanca, Guanajuato, Mexico

Computer Systems Engineering – Graduated Dec 2025 GPA: **9.4/10.0**. Focus: software architecture, databases, AI/ML, computer vision, and full-stack development.